# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **Python** and **TypeScript/Node.js** systems: durable LLM job orchestration, ETL and **Databricks** pipelines, **RAG** over vector databases, payment integrations, and ML models evaluated against baselines.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering, Object-Oriented Programming

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Built durable async **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput under concurrency without cascading task failures.
- Designed **ETL** pipelines from FastAPI services into PostgreSQL, SQL Server, and cloud storage with incremental loads and reconciliation, running 5K+ monthly jobs at 99%+ reliability across multiple destinations.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Built production **LLM pipelines** chaining answer generation, rubric generation, grading, and feedback across GPT-4o, Claude, and Qwen-VL with OCR, serving 10K+ daily requests at 90%+ grading consistency.
- Implemented model routing with timeout fallback and per-call token **cost logging** in normalized MySQL tables, plus concurrent batch ingestion processing 1K+ files per run with fault-tolerant execution.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles, CA*

- Trained an offline-**RL** Decision Transformer in PyTorch on features engineered from 100K+ trajectory logs and HD-map vectors through reproducible Hydra and W&B pipelines, improving long-horizon planning 18% in simulation.
- Built an **evaluation** loop comparing a spatial-temporal attention encoder against three baselines in closed-loop simulation, using results to select the HD-map-aware design that improved planning stability 15%.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern – VibrantChem* *India*

- Built eight **TypeScript** and Node.js microservices with React and MongoDB, synchronizing 2M+ records across **distributed** locations through REST APIs and real-time WebSocket events with allocation conflict resolution.
- Shipped Docker builds through Jenkins **CI/CD** with unit, integration, and end-to-end tests and rolling zero-downtime releases, monitoring with Sentry and Azure Monitor and reaching 95%+ build validation.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Integrated a PCI-compliant multi-currency **payment** infrastructure across cards, UPI, wallets, and PayPal with fraud checks and idempotent webhooks for refunds, disputes, and chargebacks, cutting manual payment handling 70%.
- Paired Firebase **real-time** booking sync with hourly Databricks and Spark micro-batches over 50K+ events, combining streaming state updates and **batch analytics** behind sub-200ms Python Flask booking APIs.

## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Java
**AI & Agents:** Claude, GPT-4o, LLM Pipelines, RAG, Prompt Chaining, Embeddings, Model Evaluation
**ML:** PyTorch, Transformers, Offline RL, Recommendation Systems, Feature Engineering, Hydra, W&B
**Data Engineering:** Databricks, Spark, ETL/ELT, Snowflake, AWS Glue, Batch and Real-Time Pipelines, Pinecone, FAISS
**Backend & Distributed:** Node.js, Express.js, FastAPI, Flask, PostgreSQL, MongoDB, Firebase, WebSockets, Webhooks, Async Job Orchestration
**Platform:** Payments Integration, Docker, AWS, Azure, GCP, CI/CD, Sentry

## Projects

**Agent Context and Memory Platform** | *Python, TypeScript, MCP, Temporal, Redis Streams, PostgreSQL, Qdrant*

- Designed a Python and TypeScript **MCP server** exposing agent memory and tools, with Temporal-orchestrated enrichment workflows, event-sourced Redis streams into Postgres and Qdrant, and a human-annotated **eval** set.

**Big-Brain** Live | *LLaMA-2, Pinecone, FAISS, PostgreSQL, GCP*

- Designed vector search with **Pinecone** and FAISS over transformer embeddings for a LLaMA-2 RAG platform, with incremental ingestion and metadata persistence cutting retrieval latency 40% at 95% query accuracy.

**Trend Tracker with Fashion Recommendation System** Live | *CNN, React.js, Strapi, AWS EC2*

- Built a CNN-based **recommendation engine** for an e-commerce trend tracker on React, Strapi, and AWS EC2, improving accuracy from 78% to 92% and engagement across 3M+ users.